

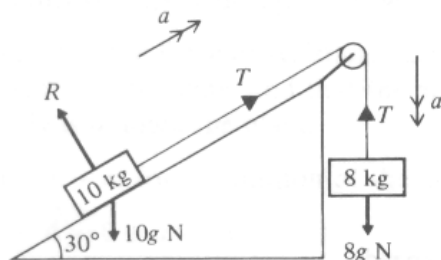
INCLINED PLANES WITH PULLEYS

12/01/26

Draw diagrams, even if you are provided with them

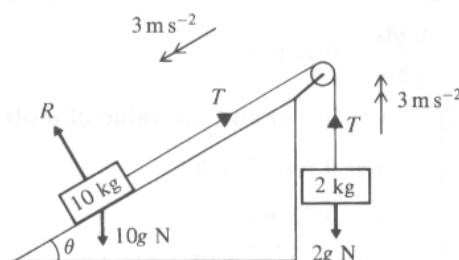
- Find the unknown quantities.

(a)



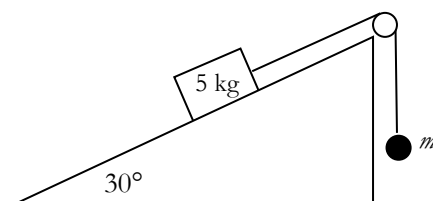
Find T , a and R

(b)



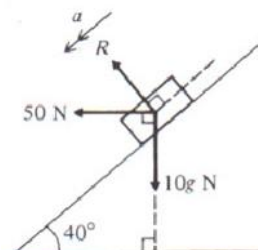
Find θ , R and T

- A block of mass 5 kg is at rest on a plane which is inclined at 30° to the horizontal. A light, inelastic string is attached to the block, passes over a smooth pulley and supports a mass m which is hanging freely. The part of the string between the block and pulley is parallel to the line of greatest slope of the plane. The frictional force between the block and the plane is 14.7 N.



- Calculate the value of m if the block is to slide up the slope at constant speed.
- Calculate the acceleration when $m = 6$ kg and find the tension in the string.

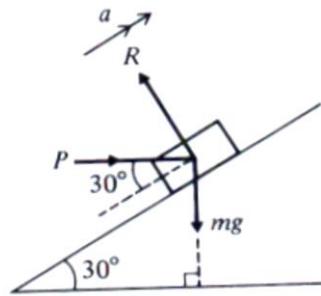
- For the system shown, where a 50 N force acts on the body horizontally, find:
 - the acceleration
 - the normal reaction R



- A body of mass 20 kg is released from rest at the top of a rough slope which is inclined at 30° to the horizontal. Six seconds later the body has a velocity of 21 m s^{-1} down the slope. Find the resistance to motion experienced by the body. (Assume this resistance to be constant throughout).
- A mass of 10 kg lies on a smooth plane which is inclined at θ to the horizontal. The mass is 5 m from the top, measured along the plane. One end of a light inextensible string is attached to this mass; the string passes up the line of greatest slope and over a smooth pulley fixed at the top of the slope. The other end is attached to a freely suspended mass of 15 kg. This 15 kg mass is 4 m above the floor. The system is released from rest and the string first goes slack $1\frac{3}{7}$ s later. Find the value of θ .

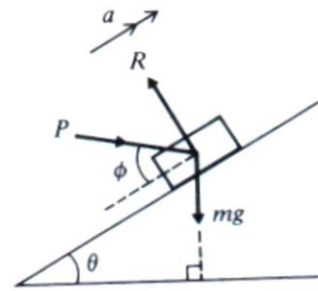
6. Prove the statements below.

(a)



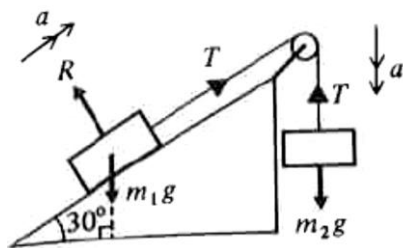
Prove: $\sqrt{3} P = m(2a + g)$

(b)



Prove: $R = m(g \cos \theta + a \tan \phi + g \tan \phi \sin \theta)$

(c)



Prove: $a = \frac{(2m_2 - m_1)g}{2(m_1 + m_2)}$